Program - 4

Detection and Counting missing values in a dataset and evaluation of data quality

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Aim

Write a Python program to detect and count missing values in a dataset and evaluate data quality using Pandas functions.

Algorithm

1. Start
2. Import the pandas and numpy libraries.
3. Create a dictionary containing student details such as:
    1. Student ID
    2. Name
    3. Department
    4. Marks
    5. Attendance
4. Convert the dictionary into a Pandas DataFrame named df.
5. Display the original dataset.
6. Check for missing values using df.isnull().
7. Calculate the number of missing values in each column using df.isnull().sum().
8. Calculate the missing-value percentage for each column.
9. Calculate the total number of missing values in the dataset.
10. Check for duplicate records using df.duplicated().sum().
11. Display the data types of all columns.
12. Find the number of unique values in each column using df.nunique().
13. Calculate the total number of cells in the DataFrame.
14. Calculate the overall missing-value percentage.
15. Calculate the Data Quality Score by subtracting the overall missing percentage from 100.
16. Classify the data quality:
    - If score = 95%, display Excellent.
    - If score = 80%, display Good.
    - If score = 60%, display Average.
    - Otherwise, display Poor.
17. Display the complete Data Quality Report.
18. Stop.

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```
import pandas as pd
import numpy as np

  data = {
        'Student_ID': [101, 102, 103, 104, 105, 106],
        'Name': ['Arun', 'Bala', 'Cathy', 'Divya', 'Ezhil', 'Fathima'],
        'Department': ['CSE', 'ECE', 'CSE', None, 'ECE', 'CSE'],
        'Marks': [85, 72, np.nan, 65, 88, np.nan],
        'Attendance': [90, np.nan, 85, 75, 92, np.nan]
       }

df = pd.DataFrame(data)
print("Original Dataset:")
print(df)
print("\nMissing Value Status:")
print(df.isnull())
missing_count = df.isnull().sum()

print("\nMissing Values in Each Column:")
print(missing_count)
missing_percentage = (df.isnull().sum() / len(df)) * 100
print("\nMissing Value Percentage:")
print(missing_percentage)
total_missing = df.isnull().sum().sum()
print("\nTotal Missing Values:", total_missing)
duplicate_count = df.duplicated().sum()
print("\nNumber of Duplicate Records:", duplicate_count)
print("\nData Types:")
print(df.dtypes)
print("\nNumber of Unique Values in Each Column:")
print(df.nunique())

total_cells = df.shape[0] * df.shape[1]
missing_percentage_overall = (
        total_missing / total_cells
) * 100

quality_percentage = 100 - missing_percentage_overall
print("\n----- DATA QUALITY REPORT ---- ")
print("Total Rows:", df.shape[0])
print("Total Columns:", df.shape[1])
print("Total Cells:", total_cells)
print("Total Missing Values:", total_missing)
print("Overall Missing Percentage:",
```

```
          round(missing_percentage_overall, 2), "%")
print("Data Quality Score:",
          round(quality_percentage, 2), "%")

if quality_percentage >= 95:
       print("Data Quality: Excellent")
elif quality_percentage >= 80:
       print("Data Quality: Good")
elif quality_percentage >= 60:
       print("Data Quality: Average")
else:
     print("Data Quality: Poor")
```